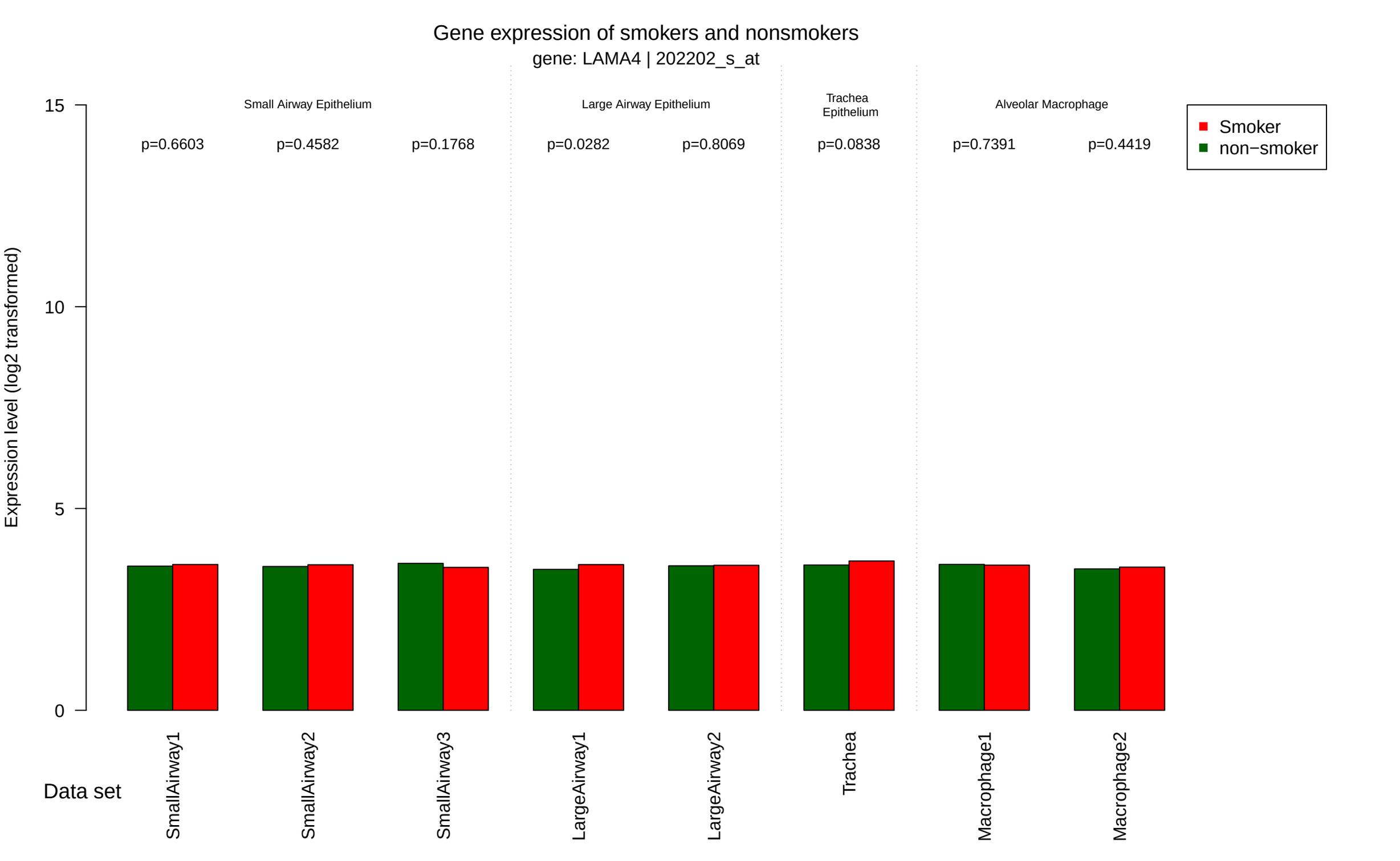
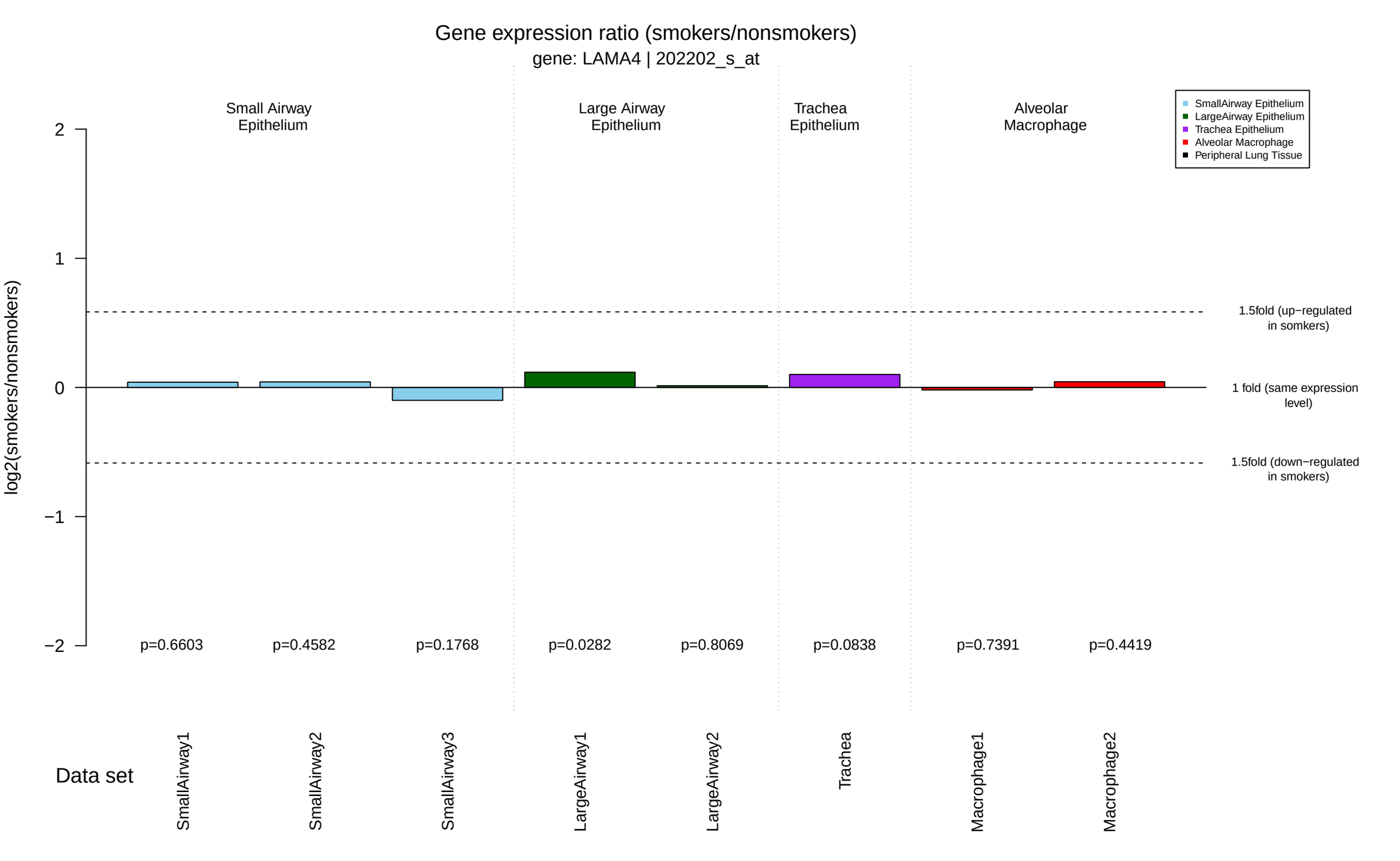
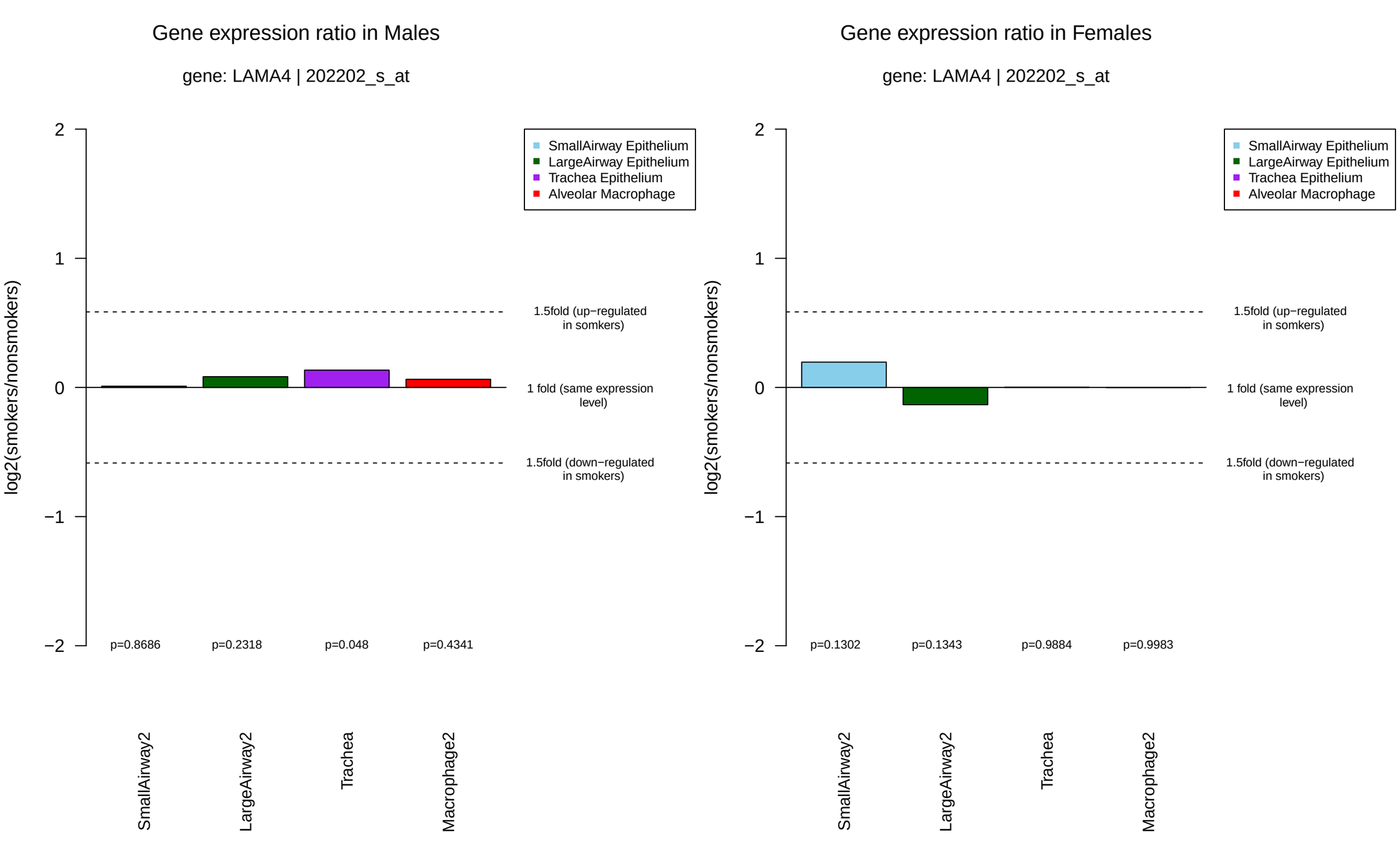
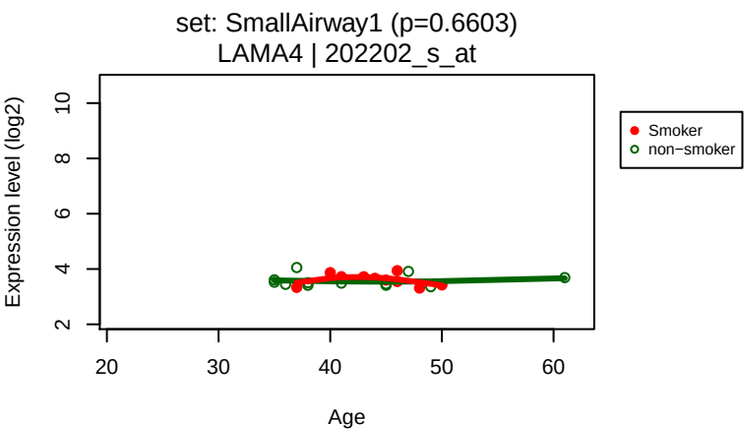
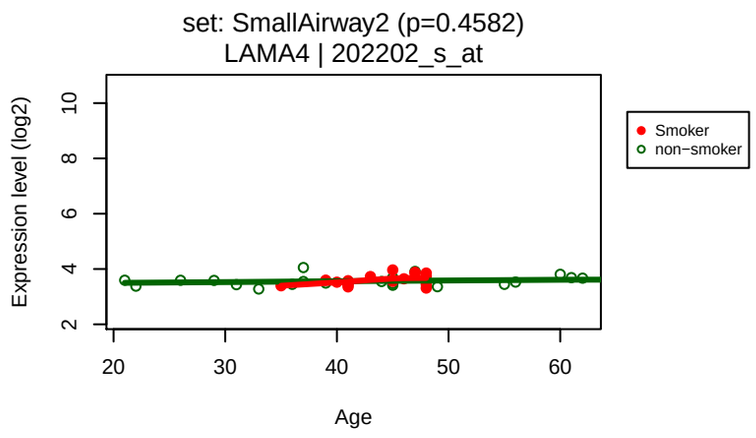
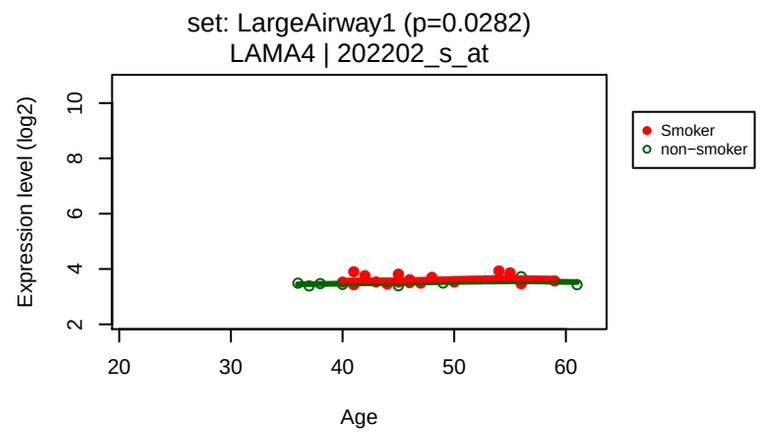

Gene expression of smokers and nonsmokers
gene: LAMA4 | 202202_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6603
p=0.4582
p=0.1768
p=0.0282
p=0.8069
p=0.0838
p=0.7391
p=0.4419
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

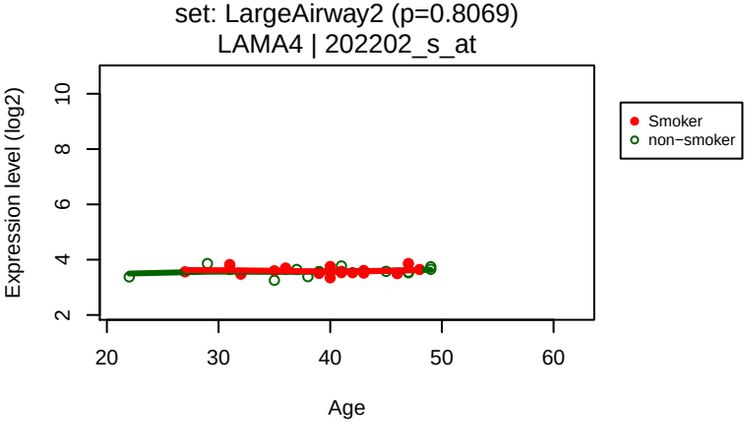

Gene expression ratio (smokers/nonsmokers)
gene: LAMA4 | 202202_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6603
p=0.4582
p=0.1768
p=0.0282
p=0.8069
p=0.0838
p=0.7391
p=0.4419
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

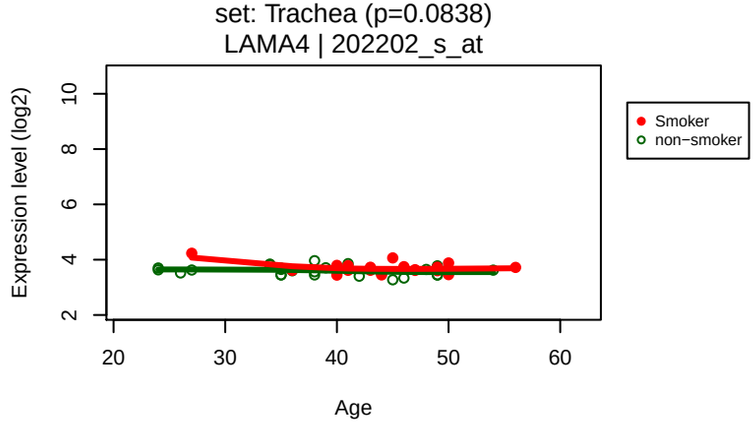

Gene expression ratio in Males
gene: LAMA4 | 202202_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8686
p=0.2318
p=0.048
p=0.4341
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LAMA4 | 202202_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1302
p=0.1343
p=0.9884
p=0.9983
SmallAirway2
LargeAirway2
Trachea
Macrophage2

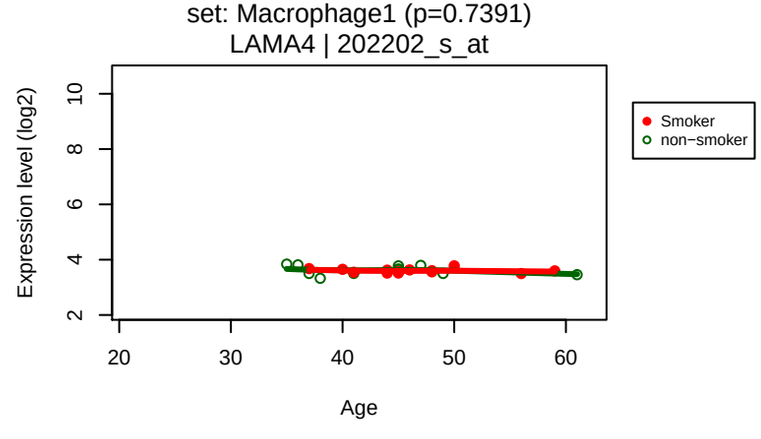

set: SmallAirway1 (p=0.6603)
LAMA4 | 202202_s_at
Expression level (log2)
Age
Smoker
non−smoker

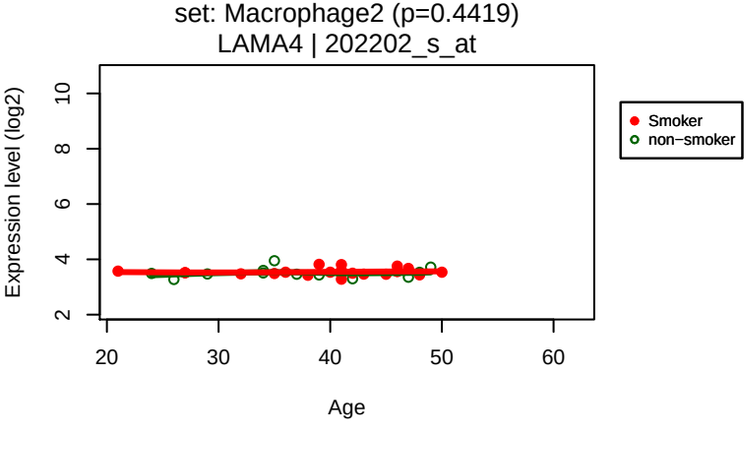

set: SmallAirway2 (p=0.4582)
LAMA4 | 202202_s_at
Expression level (log2)
Age
Smoker
non−smoker

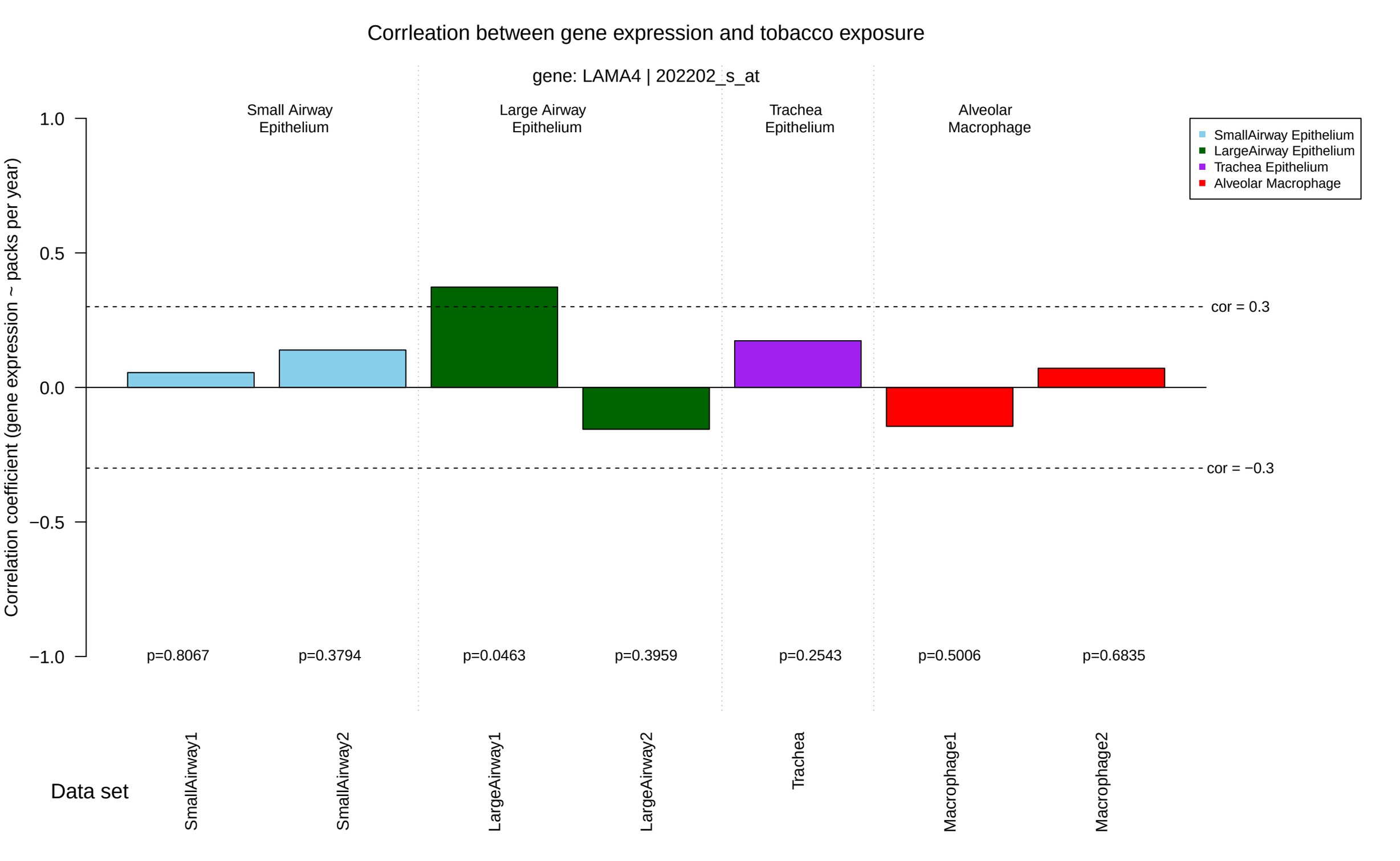

set: LargeAirway1 (p=0.0282)
LAMA4 | 202202_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8069)
LAMA4 | 202202_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0838)
LAMA4 | 202202_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7391)
LAMA4 | 202202_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4419)
LAMA4 | 202202_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LAMA4 | 202202_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8067
p=0.3794
p=0.0463
p=0.3959
p=0.2543
p=0.5006
p=0.6835
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2